

CHEM 427/527 · SESSION 3 · THURSDAY, OCTOBER 8, 2026

Reproducible practice

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Say which Python environment runs a notebook, and fix a `ModuleNotFoundError` by choosing the course kernel, not by installing a package.
2. Update your copy of the course with `git pull`, and keep your work when git refuses to pull.
3. Make a notebook run from top to bottom after a restart, with a relative path, all imports at the top, cells in the order they run, and a version cell.
4. Write a judgment entry whose reason cites a number from your own output.

FIRST, EVERYONE PULLS

Get session 3 with git pull

1. See what changed

```
git status
```

Your S1 and S2 notebooks are listed as modified. That is expected.

2. Bring in session 3

```
git pull
```

A new folder appears:

```
S03-reproducible-practice
```

We never change an activity notebook after its session.

IF GIT REFUSES TO PULL

Keep your work, then pull

What git prints

Your local changes to the following files would be overwritten by merge

1. Copy

In VS Code, copy and paste the named file. Your work is now in **activity copy.ipynb**.

2. Restore

```
git restore <path>
```

With the path git printed.

3. Pull again

```
git pull
```

The course version is up to date beside your copy.

Not git reset --hard from a web search. It deletes your work.

TODAY'S NOTEBOOK

A notebook that ran once

54.5 μ M

Iron in the unknown, from the saved output of a labmate who graduated.

Would you report it?

First read the numbers in square brackets beside each code cell.

Then click **Restart** and **Run All**.

WHICH PYTHON RUNS THE NOTEBOOK

One environment for everyone

The packages

`pyproject.toml`

Lists the packages the course uses.

The versions

`uv.lock`

Pins each one, such as NumPy 2.5.3.

The environment

`uv sync`

Builds **.venv** from both. With conda, it is **signals**.

ModuleNotFoundError?

Click the kernel name at the top right and choose **.venv**. Do not run pip install.

Four habits that survive a restart

1. Relative paths

```
data/fe_phen_calibration.csv
```

Without it: FileNotFoundError on every other laptop.

2. Imports at the top

```
import matplotlib.pyplot as plt
```

Without it: NameError: name 'plt' is not defined.

3. Cells in the order they run

A cell uses only what the cells above it define.

Without it: NameError after a restart.

4. A version cell

The last code cell prints Python and package versions.

No error today. Next year it shows whether new software changed a number.

The judgment cell

(a) Your answer

What concentration do you report for the unknown, and why?

Cite numbers from your own output.

(b) One judgment entry

- What the assistant suggested
- What I did: changed, rejected or verified
- Why, from my data

Practice today. Scored 0 to 2 from session 4. You write it yourself.

BEFORE YOU LEAVE

Checkpoint, then submit

Checkpoint

Fill in the setup survey on Canvas before you go.

Not running yet?

Agree a time with Matt, Lucas or Chris.

Every laptop works by **Tuesday, October 13.**

Submit

Upload by **1:30 pm Tuesday, October 13.**

3 if it runs after a restart, **0** if not.